



CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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0580/23

May/June 2022

1 hour 30 minutes

You must answer on the question paper.

You will need: Geometrical instruments

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should use a calculator where appropriate.
- You may use tracing paper.
- You must show all necessary working clearly.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- For π , use either your calculator value or 3.142.

- The total mark for this paper is 70.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

- 1 The probability of picking a red sweet from a bag is 0.05 .

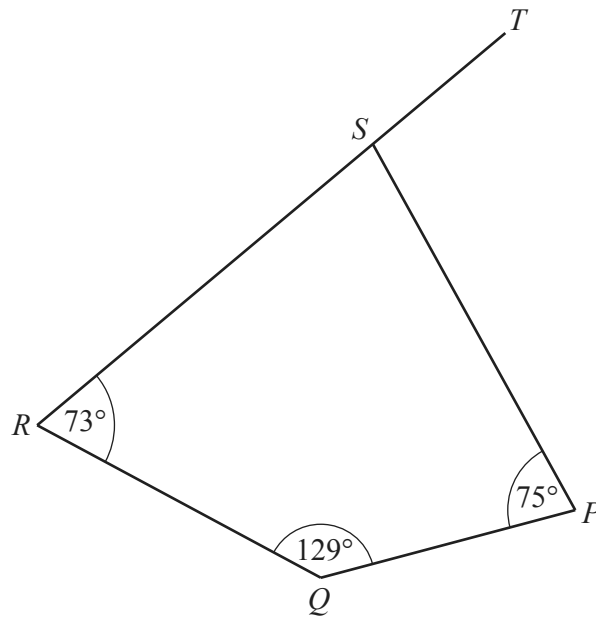
Find the probability of not picking a red sweet.

..... [1]

- 2 Work out the value of $\frac{m k^3}{\sqrt{3}}$ when $m = 4$ and $k = 7$.

..... [2]

3



NOT TO
SCALE

$PQRS$ is a quadrilateral.
 RST is a straight line.

Find angle PST .

Angle $PST =$ [2]

- 4 These are the masses, in kg, of 12 parcels.

0.3 0.4 1.2 0.8 1.1 2.1 1.7 1.8 1.2 2.3 0.7 1.1

- (a) Complete the stem-and-leaf diagram for the 12 parcels.

0	3 4
1	
2	

Key: 0 | 3 represents 0.3 kg

[2]

- (b) Find the median.

..... kg [1]

- 5 The n th term of a sequence is $n^2 - 1$.

Find the first three terms of this sequence.

.....,, [2]

- 6 Simplify.

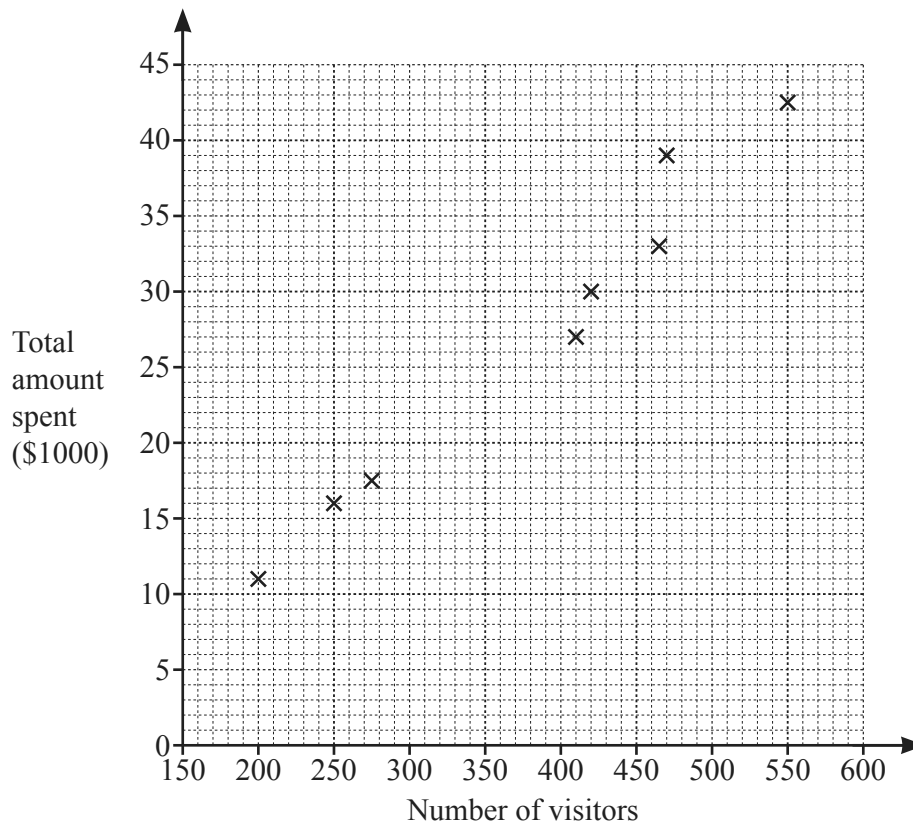
(a) $y^3 \div y^5$

..... [1]

(b) $7x^0$

..... [1]

- 7 The scatter diagram shows the number of visitors and the total amount spent, in thousands of dollars, at a zoo on each of eight days.



- (a) On one of the eight days there are 410 visitors.

Find the total amount spent by visitors during this day.

\$ [1]

- (b) Information for the ninth day is shown in the table.

Number of visitors	175
Total amount spent (\$1000)	9

Plot this information on the scatter diagram.

[1]

- (c) Draw a line of best fit on the scatter diagram.

[1]

- (d) On the tenth day the total amount spent is \$22 000.

Estimate the number of visitors on this day.

..... [1]

- 8 Without using a calculator, work out $\frac{2}{9} \div \frac{5}{6}$.

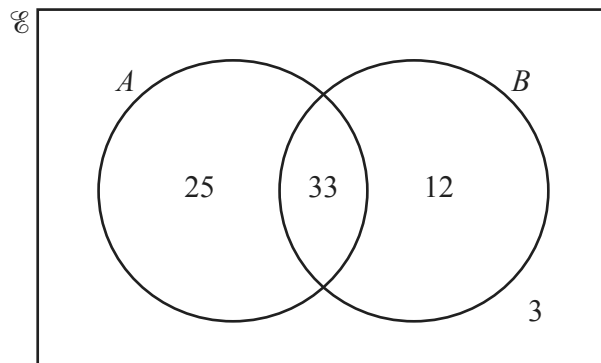
You must show all your working and give your answer as a fraction in its simplest form.

..... [2]

- 9 Change 300 m/min to km/h.

..... km/h [2]

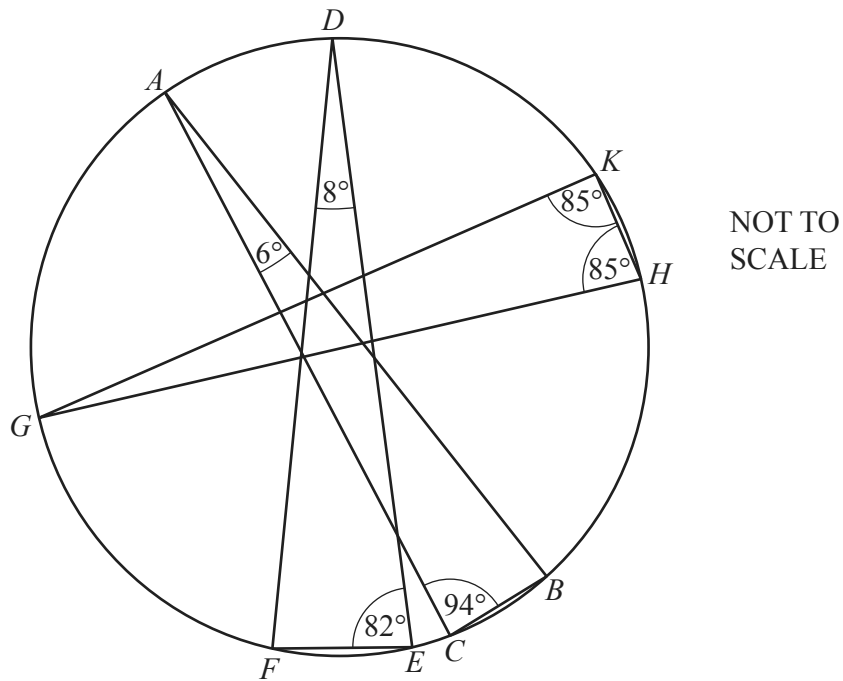
10



Find $n(A \cap B)$.

..... [1]

- 11 ABC , DEF and GHK are triangles with all vertices on the circumference of a circle.



From the list, draw a ring around the line that is a diameter of the circle.

AB AC DE DF GH GK

[1]

- 12 f is a common factor of 14 and 28.
 m is a common multiple of 10 and 25.
 p is a prime number.

Work out the largest possible value of $\frac{f}{mp}$.

..... [4]

13 Factorise completely.

(a) $18px - 27p$

..... [2]

(b) $mt - n - m + nt$

..... [2]

14 Find the n th term of this sequence.

8, 17, 32, 53, 80, ...

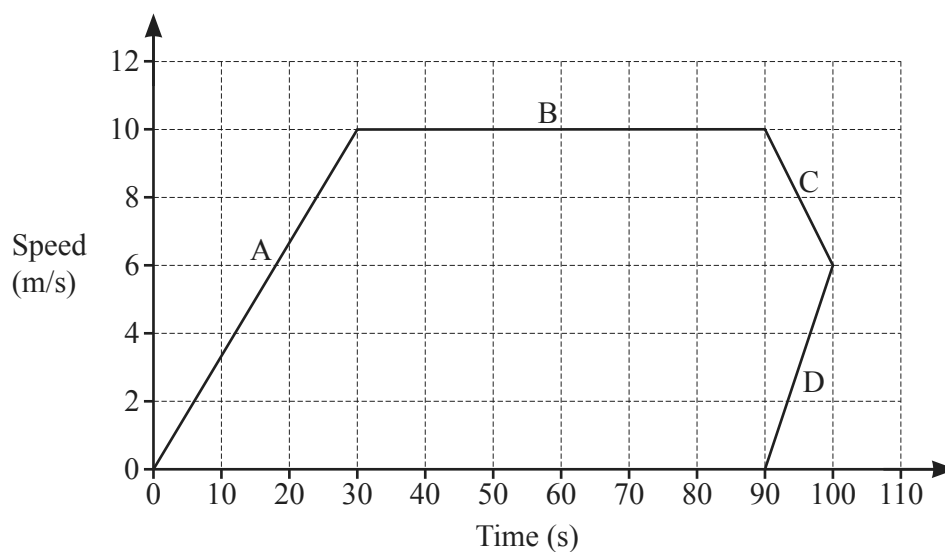
..... [2]

15 Solve.

$12x - 3 \geq 4x + 13$

..... [2]

- 16 Abdul draws this speed–time graph for a journey. The graph has four sections A, B, C and D.



Complete these statements about the speed–time graph.

Section cannot be correct.

Section shows constant speed.

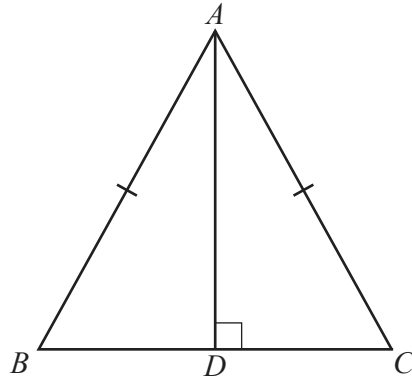
Section shows deceleration.

Section A shows acceleration of m/s^2 .

The distance travelled in the first 30 seconds of the journey is m.

[4]

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SCALE

In triangle ABC , $AC = AB$.

D is the point on BC such that AD is perpendicular to BC .

Complete the following statements to show that triangle ACD and triangle ABD are congruent.

AD is perpendicular to BC so that Angle = Angle = $^\circ$

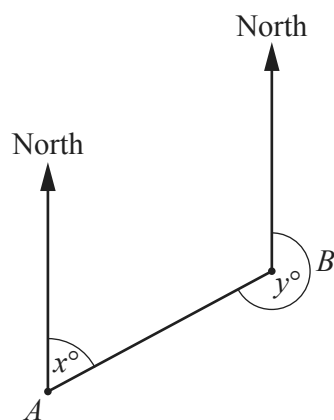
$AC = AB$ is given information.

Side is common to both triangles.

Triangle ACD is congruent to triangle ABD because of the congruency criterion [3]

- 18 The bearing of B from A is x° .
The bearing of A from B is y° .
 $x : y = 2 : 7$

Calculate the value of y .



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$y =$ [3]

19

$$f(x) = kx^2$$

$$g(x) = \frac{1}{x}$$

$$h(x) = \frac{7x-2}{5}$$

$$j(x) = \frac{3-10x}{14}$$

(a) $f(-5k) = 675$

Find the value of k .

$$k = \dots\dots\dots [2]$$

(b) Find $gh(x)$.

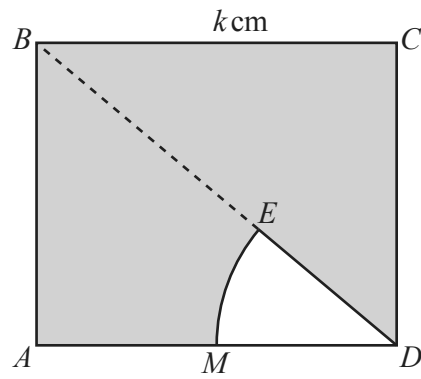
$$\dots\dots\dots [1]$$

(c) Find $h^{-1}(x) + j(x)$.

Give your answer in its simplest form.

$$\dots\dots\dots [4]$$

20

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The diagram shows a square $ABCD$ with side length k cm.
 MDE is a sector of a circle, centre D .
 E lies on the diagonal, BD , of the square.
 M is the midpoint of AD .

Find the percentage of the square that is shaded.

..... % [4]

- 21 Neha has a piece of ribbon of length 23 cm, correct to the nearest cm.
From this ribbon she cuts off a piece with length 87 mm, correct to the nearest mm.

Work out the lower bound and the upper bound for the length of the remaining ribbon.
Give your answer in centimetres.

Lower bound = cm

Upper bound =cm [3]

- 22 Simplify.

$$\frac{5x - x^2}{25 - x^2}$$

..... [3]

23 Solve the equation $3 \sin x + 3 = 1$ for $0^\circ \leq x \leq 360^\circ$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ [3]

24 y is inversely proportional to the cube of $(x - 1)$.
 $y = 9.45$ when $x = 3$.

Find y when $x = 4$.

$y = \dots\dots\dots$ [3]

25 $m^{-\frac{1}{4}} = 27m^{-1}$

Find the value of m .

$m = \dots\dots\dots$ [3]

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